DINESH KUMAR S

AI Engineer • Generative AI & Agentic Systems • NLP & RAG Specialist

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PROFESSIONAL SUMMARY

Results-driven Software Developer with 2 years 1 month of hands-on experience designing and developing production-grade Generative AI, LLM, and Agentic AI systems for large-scale financial document intelligence. Proven expertise in RAG architectures, NLP extraction pipelines, multi-modal document processing, and LangGraph-based multi-agent orchestration. Skilled at bridging ML research and real-world engineering — from semantic embeddings and vector search to automated agentic workflows using n8n and MCP integrations. Experienced with local LLM deployment using Ollama (Qwen, LLaMA) for private, cost-effective inference. Passionate about building reliable, grounded AI systems that reduce hallucinations and drive measurable business value.

TECHNICAL SKILLS

AI / GenAI:	LLMs, Retrieval-Augmented Generation (RAG), Agentic AI, Prompt Optimization, Semantic Search, Vector Databases (FAISS, Chroma), Hallucination Reduction, AI Agents, Tool Calling , FineTunning , Vector-Less RAG
Agentic / Workflow:	LangGraph, n8n Automation, MCP (Model Context Protocol), Multi-Agent Orchestration, ReAct Agents, Autonomous Pipelines, Multi-modal AI Agents, LangServe .
Local LLM / Ollama:	Ollama, Qwen3-Coder:30b, Qwen2.5:7b, LLaMA 3.1, Local Inference, Agent-based Field & Table Extraction, Quantized Model Deployment, AI Observability
ML /DL/ NLP:	Natural Language Processing, TF-IDF, Text Classification, Named Entity Recognition (NER), Embeddings, Image Processing, OCR, CNN , Pytorch , TensorFlow , keras
Frameworks:	LangChain, LangGraph, LlmaIndex, FastAPI, Streamlit, n8n, Hugging Face, Scikit-learn
Models:	LLaMA-70B, Mistral, GPT-4 / OpenAI, Qwen3-Coder, Qwen2.5, ResNet, VGG16, Vision Transformers (ViT) , Gemini flash
Languages:	Python, SQL
Tools & Cloud:	Microsoft Azure, Power BI, Git, Docker, EDA , Linux, Vector Databases, IDP

WORK EXPERIENCE

Software Developer — Data Mining & AI | *Factentry Data Solutions Pvt. Ltd. (SIX Group)* Jun 2025 – Present

- Architected and deployed ML-driven NLP extraction pipelines and RAG systems using semantic embeddings and vector search, enabling accurate retrieval-augmented generation over large-scale financial PDFs.
- Implemented query-time retrieval mechanisms to ground LLM responses, significantly reducing hallucinations and improving output accuracy in production environments.
- Built Agentic AI workflows on the n8n automation platform to orchestrate multi-step decision-making, automated email reporting, and intelligent content generation.
- Designed LangGraph-based multi-agent systems for complex, stateful document workflows — enabling conditional routing, parallel agent execution, and iterative self-correction.
- Deployed local Ollama agents (Qwen3-Coder:30b, Qwen2.5:7b) for private tabular and field-level extraction, eliminating cloud API costs for sensitive financial data.
- Designed end-to-end GenAI-powered document intelligence solutions integrating OCR, LLMs, and semantic search for structured financial data extraction at scale.

Graduate Intern — Data Science & MLOps | *Factentry Data Solutions Pvt. Ltd. (SIX Group)* Nov 2024 – May 2025

- Engineered multilingual PDF analysis pipelines integrating OCR, regex, language detection, and LLMs — supporting structured extraction from noisy, real-world financial documents.
- Developed advanced image-based document clustering using ResNet, VGG16, and Vision Transformers (ViT), combined with PCA + DBSCAN and 3D visualization for document triage.
- Built a scalable multi-modal PDF Q&A chatbot (FastAPI + Streamlit + Groq LLaMA-70B + LangChain) with NER, conversational memory, and page-level citation support.

KEY PROJECTS

ML-Driven NLP Extraction System

- Built TF-IDF and probabilistic ML classifiers for high-precision unstructured document extraction with sliding-window context analysis and adjacent-sentence merging for noisy text.
- Deployed serialized vectorizers and classifiers into a hybrid ML + rule-based production inference pipeline with confidence-based decision thresholds.
- **Tech Stack:** Scikit-learn, NLTK, spaCy, Regex, Python

Generative-AI Conversational Agent for Fixed Income Analysis

- Developed a conversational PDF Q&A agent integrating Mistral and LLaMA-70B (via Groq API) with Streamlit for fixed income document analysis.
- Implemented table extraction, metadata analysis, multilingual summarization, page-level referencing, NER, and conversational memory using LangChain and PDFPlumber.
- **Tech Stack:** LangChain, Groq, Streamlit, PDFPlumber, NLTK, Prompt Engineering

Generative-AI-Powered RAG-Based Document Extraction System

- Designed a production-grade RAG pipeline transforming unstructured financial PDFs into structured, analytics-ready data using semantic chunking, embedding generation, and FAISS vector search.
- Architected context-aware retrieval with grounded LLM responses; integrated Agentic AI for multi-step reasoning, validation, and field-level extraction at scale.
- **Tech Stack:** LangChain, FAISS, OpenAI / Groq LLMs, FastAPI, Python

Agentic AI Email Reporting Automation (n8n + MCP)

- Designed an end-to-end autonomous email reporting system on the n8n platform with multi-step AI agents that analyze data, generate contextual reports, and dispatch personalized emails.
- Integrated LLMs, external APIs, and MCP tool servers within n8n to enable intelligent summarization, validation, and scheduled delivery — fully automated, zero human intervention.
- **Tech Stack:** n8n, LLM APIs, MCP, LangGraph, GenAI, Workflow Automation

Agentic AI-Based Financial Document Intelligence

- Built a 7-agent AI workflow for financial document extraction, validation, arbitration, calculation, and auditing using LangGraph and CrewAI.
- Developed Hybrid RAG with dense embeddings, BM25, RRF, and cross-encoder reranking for evidence-based retrieval.
- Integrated Gemini LLMs with Regex/ML extractors to extract structured bond data with source evidence and JSON outputs.
- Implemented AI guardrails and reward-based arbitration for hallucination detection, extraction conflict resolution, and feedback-driven source selection.
- Automated cross-field validation, audit-based retries, and Excel reporting with provenance and confidence metadata.
- **Tech Stack:** Python, Agentic AI, Multi-Agent Orchestration, RAG, Prompt Engineering, LLM Integration, Hybrid Search, AI Guardrails, Reward-Based Learning, Automated Evaluation, Python Data Pipelines.

EDUCATION

M.Tech Integrated — CSE, Specialization: Data Science 2025
Vellore Institute of Technology (VIT), Vellore | CGPA: 7.68 / 10

CERTIFICATIONS

Microsoft Azure AI Fundamentals — Microsoft | 2023
Applied Data Science — Smart Internz | 2023